

Matched-Magnitude Controls for Causal Testing of Spectral Mechanisms in Attention

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Abstract

Mechanistic claims about the spectral structure of attention — that some eigenmode, gap, or spectral direction *carries computation* — are increasingly common, but they are rarely subjected to a causal test, and when they are, the test often lacks a control for the magnitude of the intervention. We argue that this omission is not a detail: an ablation that suppresses a “dominant” spectral mode almost always perturbs the model by a larger amount than a naive control, and that magnitude asymmetry alone produces a downstream effect that is easily mistaken for evidence of a computational mechanism. We give a worked example from our own work in which a natural intervention construction produced a treatment perturbation 10–30× larger than its control — an asymmetry that would have read as a strong positive result — and was caught only by a matched-magnitude sanity check before the full run. We then describe a simple framework — matched-magnitude intervention, a random-spectrum baseline, and pre-registered outcome criteria — and report that, once the magnitudes were matched, the same hypothesis returned a pre-registered null. The note is intended as a short, reusable checklist for anyone running causal tests of spectral mechanisms.

1 The problem

A recurring pattern in the interpretability of attention is the spectral mechanistic claim: an eigenvalue, a spectral gap, a dominant eigenvector, or a low-rank direction of some attention-derived operator is identified, shown to correlate with model behaviour, and then described as a mechanism — something the model *uses* to compute. The spectral object is real and measurable; the inferential step from “measurable and correlated” to “computationally load-bearing” is where the difficulty lies.

Correlational evidence cannot settle that step. A spectral quantity can be perfectly reproducible, mathematically well-defined, and tightly coupled to other quantities, and still be causally inert — a description of the computation rather than a part of it. Deep learning offers many such structures: embedding anisotropy, neuron clusters, loss-landscape valleys. Each is real; none is, on its own, a mechanism. Distinguishing the two requires intervention: change the spectral object and measure whether the computation changes.

Intervention, however, introduces its own failure mode, and it is this failure mode that the present note is about. An intervention is only informative relative to a control, and the control must be matched in the right variable. The variable that matters is the *magnitude of the perturbation applied to the model*. If the treatment — ablating the spectral object of interest — perturbs the model more than the control, then any downstream difference is confounded: it may reflect the special role of the object, or it may reflect nothing more than “a larger perturbation does more damage.” These

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two explanations are not distinguishable after the fact. They must be separated by construction, before the experiment is run.

This is easy to state and easy to violate, because the natural constructions of a spectral intervention do not preserve perturbation magnitude. We show this concretely next.

2 A worked example

We take the example from our own work, because we have the full record of what went wrong and how it was caught.

The hypothesis. The free-energy Hessian of an attention head can be written $H = -\beta \text{Cov}_p(k)$, where k are the key vectors, p are the softmax attention weights of a query, and β is the scaled-dot-product temperature. On trained models this Hessian is typically spike-dominated: one eigenvalue λ_1 is much larger in magnitude than the rest. The hypothesis under test was that this dominant mode is a privileged computational object — that the model relies on it, and that removing it should selectively damage next-token prediction.

The intervention. To test this causally, one ablates the dominant mode and measures the downstream effect. The mode is a direction v_1 in key space; ablating it means suppressing the variance of the keys along v_1 . The control, correspondingly, suppresses a random non-dominant direction v_j from the bulk of the spectrum. If the dominant mode is special, the treatment must hurt *more* than the control.

The naive construction and its hidden asymmetry. The first, natural construction was: project the key matrix to remove its component along the chosen direction, then rescale the result to preserve the Frobenius norm $\|K\|_F$. Preserving $\|K\|_F$ feels like the right invariant — the keys keep their overall scale — and it is a quantity one naturally thinks to hold fixed.

It is the wrong invariant. Preserving $\|K\|_F$ does not preserve the magnitude of the *perturbation* $\|\delta K\|_F$, and the perturbation magnitude is what the control must match. Removing the dominant direction v_1 deletes a large component of the keys; removing a bulk direction v_j deletes a tiny one. Rescaling to a common $\|K\|_F$ afterwards does not undo this — it leaves the treatment having moved the keys much farther than the control. In our runs the dominant-mode projection produced a perturbation $\|\delta K\|_F$ that was 10 to 30 $\times$ larger than the bulk-mode control, depending on the layer-head pair. Concretely, for one early-layer head the dominant direction carried a key-projection magnitude $M_T \approx 84.9$ against the bulk direction’s $M_C \approx 15.8$ — a factor of 5.4 in magnitude, and larger once expressed as a perturbation norm.

Why this would have looked like a discovery. An uncontrolled smoke run with this construction showed the treatment producing a downstream KL divergence roughly 18 $\times$ that of the control. Taken at face value, that is a striking positive result: “the dominant spectral mode is far more important than a random mode.” It is exactly the number one would hope to see if the hypothesis were true. It is also entirely consistent with the dominant mode being computationally irrelevant — because the treatment had simply perturbed the model 10–30 $\times$ harder. The experiment, as constructed, could not tell the two apart. The asymmetry was caught not by inspecting the result — the result looked good — but by a separate sanity check that measured $\|\delta K\|_F$ for treatment and control directly, before committing to the full run.

3 Why the naive control deceives

The mechanism of the deception is worth stating precisely, because it generalises beyond this one construction.

Any intervention on a model induces a downstream change, and to first order the size of that change grows with the size of the intervention. Write the downstream effect as $E(\delta)$, a function of the perturbation δ applied. For small perturbations E is monotone increasing in $\|\delta\|$: a larger push produces a larger effect, almost regardless of *where* the push is applied. A causal test of “does direction A matter more than direction B ” must therefore compare E at *equal* $\|\delta\|$. If instead the test compares $E(\delta_A)$ against $E(\delta_B)$ with $\|\delta_A\| \gg \|\delta_B\|$, the outcome $E(\delta_A) > E(\delta_B)$ is overdetermined — it follows from the magnitude gap alone and carries no information about A versus B .

The trap is that the magnitude gap is invisible unless explicitly measured. The experimenter chooses an invariant to preserve — here, $\|K\|_F$ — believes the comparison is therefore fair, and never inspects $\|\delta K\|_F$. The preserved invariant and the invariant that actually needs preserving are different quantities, and nothing in the construction signals the difference. The result then looks like a clean positive, because the confound and the hypothesis predict the same sign.

This is a general hazard for spectral interventions specifically, because spectral objects are defined by magnitude: the “dominant” mode is, by definition, the direction along which the operator has the largest action. Ablating it is intrinsically a larger intervention than ablating a bulk mode. The asymmetry is not a bug in one construction; it is built into what “dominant” means. Any spectral causal test that does not explicitly neutralise it will tend to confirm the mechanism whether or not the mechanism is real.

4 The framework

The remedy is not elaborate. It has three components, and the value of stating them as a checklist is precisely that they are easy to omit.

1. Matched-magnitude intervention. Construct treatment and control so that the perturbation norm $\|\delta\|$ applied to the model is equal — not an indirect invariant such as the operator norm of the keys, but the perturbation itself. In our corrected construction this was done by computing, for each layer-head pair, the key-projection magnitude of both the dominant direction (M_T) and the random bulk direction (M_C), and scaling both interventions to the smaller of the two, $T = \min(M_T, M_C)$. The treatment and control then move the model by the same amount by construction. After the fix, the verified treatment-to-control magnitude ratio was 1.000 on every layer-head pair tested. Matching the perturbation has a price — in our case, exact preservation of $\|K\|_F$ had to be given up, and the resulting change in $\|K\|_F$ was logged (sub-percent for typical interventions). That trade is correct: the perturbation match is the invariant the causal claim depends on; $\|K\|_F$ is not.

2. A random-spectrum baseline. Matching magnitude controls for “how hard did we push.” A second control addresses “is the structure we are testing even non-trivial.” Before attributing significance to a spectral object, check whether a random spectrum — random eigenvalues of the same dimension, carrying no trained structure — exhibits the same property. If it does, the property is a feature of the spectral summary, not of the trained model. We found this necessary in practice: a coupling among three spectral summary statistics that looked like a low-dimensional structure of trained heads turned out to be reproduced exactly by random spectra — it was an algebraic

Table 1: Matched-control ablation of the dominant spectral mode. The operationally meaningful metric — next-token NLL — shows a treatment-minus-control differential whose 95% confidence interval brackets zero. The result is a pre-registered null.

Metric	$\bar{\Delta}_{\text{treat}}$	$\bar{\Delta}_{\text{ctrl}}$	differential	Wilcoxon p
NLL (next-token)	5.95×10^{-4}	3.23×10^{-4}	$+2.72 \times 10^{-4}$	0.26
KL (distribution)	1.31×10^{-3}	7.81×10^{-4}	$+5.32 \times 10^{-4}$	$< 10^{-38}$

identity among monotone functions of spectral peakedness, not a property of the network. A random-spectrum baseline catches this class of non-finding before it is built upon.

3. Pre-registered outcome criteria. Fix, before running, what the two outcomes are and what each means. A causal test is only honest if a null result is as reportable as a positive one. We pre-registered: if the matched treatment-minus-control effect was statistically indistinguishable from zero, the dominant-mode hypothesis was to be declared null and the direction closed; if the treatment effect significantly exceeded the control, the hypothesis would continue. Writing this down in advance removes the post-hoc freedom to reinterpret a null as “a weak positive” or to explain it away.

5 Result of applying the framework

With the three components in place, the hypothesis of Section 2 was tested on Llama-3.2-1B over WikiText-2: twelve layer-head pairs spanning early, middle, and late layers, each on twenty-five prompts, 300 measurements in total, at a context length of 256 keys. Each measurement compared three forward passes — unablated baseline, dominant-mode treatment, matched bulk-mode control — on the same prompt.

The outcome is shown in Table 1. On next-token NLL — the operationally meaningful measure — the treatment-minus-control differential is $+2.7 \times 10^{-4}$, with a 95% confidence interval of $[-1.7 \times 10^{-4}, 7.0 \times 10^{-4}]$ that brackets zero, and a Wilcoxon signed-rank $p = 0.26$. By the pre-registered criterion this is a null: suppressing the dominant spectral mode harms next-token prediction no more than suppressing a random bulk mode of the same perturbation magnitude. As a fraction of the baseline NLL the differential is $\sim 10^{-4}$ — operationally negligible, independently of the significance test. The effect did not concentrate in any layer band; the per-band differentials are individually consistent with zero.

The KL divergence between treatment and control output distributions is statistically distinguishable from zero ($p < 10^{-38}$), but its absolute scale is $\sim 5 \times 10^{-4}$ nats, and it does not translate into NLL harm. A detectable distributional shift that does not affect task performance is consistent with the dominant mode being a description of the computation rather than a load-bearing part of it.

The contrast with the uncontrolled smoke run is the point of this note. The same hypothesis, with the same model and the same kind of intervention, returned a treatment KL of roughly $18\times$ the control under the naive construction and a null under the matched construction. The difference between a reported discovery and a reported null was entirely the magnitude match. Figure 1 shows the verification that, after the fix, the treatment and control perturbation magnitudes were equal across all twelve pairs.

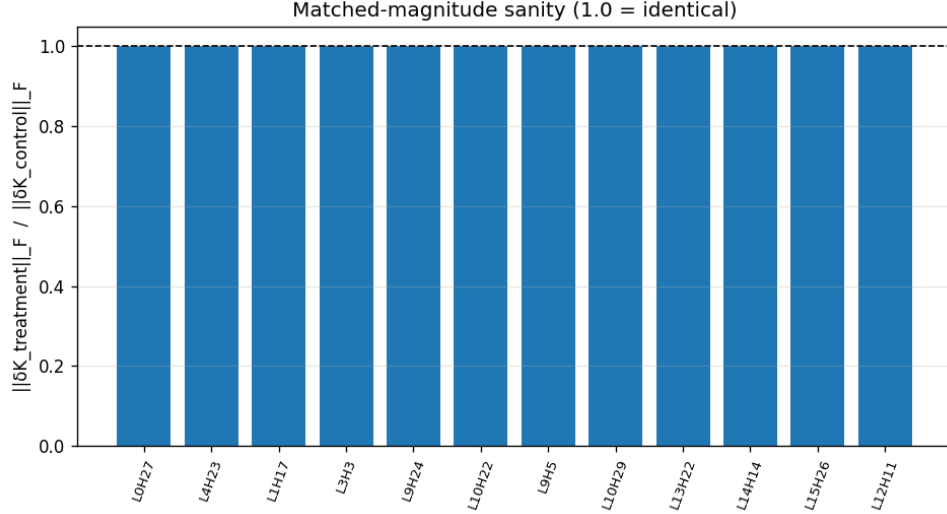


Figure 1: Matched-magnitude verification. For each of the twelve layer-head pairs, the perturbation magnitude applied in the treatment (dominant-mode ablation) and in the control (random bulk-mode ablation) after the corrected construction. The ratio is 1.000 across all pairs: the two interventions move the model by the same amount, so the downstream comparison is not confounded by perturbation size. Under the naive $\|K\|_F$ -preserving construction this ratio was 10–30.

6 Recommendation

We close with the checklist the note argues for. It is short by design; its usefulness is that each item is easy to skip and costly to skip.

1. **Match the perturbation, not a proxy.** Construct treatment and control so that the perturbation norm $\|\delta\|$ applied to the model is equal. Do not substitute an indirect invariant (operator norm, parameter norm) for the perturbation itself — they are different quantities, and only the perturbation match neutralises the magnitude confound.
2. **Measure the match; do not assume it.** Compute and report $\|\delta_{\text{treat}}\|/\|\delta_{\text{ctrl}}\|$ explicitly. A construction that looks symmetric can carry a large hidden asymmetry; the only way to know is to measure it, before the full run.
3. **Add a random-structure baseline.** Check whether a random spectrum of the same dimension reproduces the property under study. If it does, the property is algebraic, not a fact about the trained model.
4. **Pre-register both outcomes.** State, before running, what constitutes a null and what constitutes a positive, and commit to reporting a null as such. A spectral causal test is only evidence if it could have come out the other way.
5. **Separate the distributional from the operational.** A significant KL shift is not a mechanism if it does not move the task-level metric. Report both, and do not let a small significant distributional effect stand in for an operational one.

None of this is specific to attention or to the particular Hessian we studied; it applies to any causal test of a spectral object in a trained network. The reason to write it down is empirical: we adopted

exactly one of these five items late, by accident of a sanity check, and it converted what would have been a reported positive into a correct null. The other four are cheap insurance against the same class of error.

Data and code. The intervention construction, the matched-magnitude verification, and the full ablation results are available with the spectral-gap-statement research record at <https://doi.org/10.5281/zenodo.20257482>.